

RAJASIMHA MERUGUMALLA

Machine Learning / AI Engineer

+91 7416470528 | merugumallarajasimha@gmail.com | LinkedIn | GitHub

Professional Summary

Computer Science (Data Science) student with hands-on experience in AI engineering, machine learning, data analytics, and business intelligence. Proficient in Python, SQL, Power BI (DAX, Power Query, Star Schema), ETL pipeline development, exploratory data analysis, predictive modeling, and end-to-end ML deployment. Demonstrated ability to transform raw data into actionable business insights through dashboards, statistical analysis, and autonomous AI systems.

Technical Skills

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL

AI & Machine Learning: LangGraph, Qdrant Vector DB, BM25, Cross-Encoders, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Ensemble Methods, Gradient Boosting (XGBoost, LightGBM, CatBoost)

BI & Visualization: Power BI, Tableau, Excel (Pivot Tables, Advanced Functions, VLOOKUP)

Data Skills: ETL Pipelines, Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis, A/B Testing Concepts

Tools & Platforms: Jupyter Notebook, Streamlit, Docker, FastAPI, Git/GitHub, VS Code, Linux

Projects

Autonomous Multi-Agent Coder (Autocoder) | *Personal Project*

2026

- To solve manual debugging latency in software engineering workflows, architected a local-first multi-agent code generation engine using **LangGraph** and **Qwen2.5-Coder** via **Ollama**, automating repository inspection and code synthesis [[GitHub](#)].
- Engineered a secure **Docker execution sandbox** configured with isolated networking (`--network=none`) and 512MB RAM resource constraints, safely testing untrusted AI-generated Python code without host exposure.
- Implemented an automated **self-healing loop** utilizing **Ruff** and **Bandit** to detect syntax and security errors, automatically patching code flaws prior to human Git approvals.
- Designed a model fallback router that dynamically cascades from local Ollama instances to cloud LLM APIs during resource bottlenecks, ensuring 99.9% pipeline uptime.

RAGLens — Citation-Aware Document Intelligence | *Personal Project*

2026

- Built a high-precision research paper RAG pipeline combining **Qdrant** dense embeddings and **BM25** sparse search via **Reciprocal Rank Fusion (RRF, k=60)**, elevating context retrieval accuracy by 38% [[GitHub](#)].
- Incorporated **ms-marco-MiniLM-L-6-v2 Cross-Encoders** for top-*k* reranking, filtering out non-relevant context chunks and reducing downstream LLM token consumption by 25%.
- Engineered a custom post-generation **CitationValidator** that cross-checks LLM response IDs against assigned context metadata, eliminating hallucinated citation tags prior to UI rendering.
- Developed multi-pipeline intent routing in **FastAPI** to execute specialized workflows for technical QA, evidence extraction, and side-by-side literature comparisons.

TransitGuard — Predictive Delivery ML Pipeline | *Personal Project*

2026

- Constructed an end-to-end delivery cancellation and delay prediction pipeline powered by **LightGBM**, serving real-time risk scores to mitigate supply chain disruptions [[GitHub](#)].
- Resolved severe class imbalance (Class 2 cancellation rate at 4.3%) using **cost-sensitive learning** (`class_weight='balanced'`), capturing **97% recall** on high-risk cancellations.
- Architected a **PostgreSQL data warehouse** with star-schema modeling, sanitizing post-checkout data leakages to preserve feature integrity.
- Containerized real-time prediction endpoints using **FastAPI** and **Docker**, achieving sub-50ms inference latency and serving an interactive monitoring dashboard in **Streamlit**.

Education

Malla Reddy College of Engineering & Technology

B.Tech in Computer Science (Data Science)

Hyderabad, India

2022 – 2026